- After the payment is claimed, the channel is closed. A new channel is created, and the process is repeated for every worker and every session of the worker. This opening and closing of an account is impossible in a traditional banking system but is a normal process in a blockchain system.

Let’s design and develop MPC-Dapp to demonstrate how it solves these issues by using an off-chain micropayment channel. This off-chain instrument will be supported by on-chain blockchain services using a smart contract and secure digital signing. And of course, web3 API will be used to access these blockchain services for all these operations. Before you develop a blockchain-based solution, however, don’t overlook traditional financial systems, such as banks; assess whether they can solve the MPC problem without involving a blockchain.

7.4.1 Traditional banking solution

Figure 7.5 shows a possible solution that uses the conventional banking system to pay for the massive, decentralized global plastics cleanup. In this case, the organizer of MPC will deposit the escrow in the bank and somehow (such as through off-chain messaging) let the pre-identified workers know that they can start the cleanup. For this discussion, you can assume that the relationship between the organizer (sender) and a worker is one-to-one. A worker collects garbage in bins (in this case, 5, 1, and 2, as shown in figure 7.5), and keeps sending information about this collection to the organizer. In this scenario, the organizer first issues a check for $5 ($1 per bin), followed by a check for $1, and finally a check for $2.

The worker gives the bank the checks to cash in. The bank verifies the signature on the checks and pays the worker from